

Understanding microwave shielding of molecules

Huan Bui

ZGS May 23, 2025

Outline

- Anecdote
- Universal two-body loss
- (Single) microwave shielding
- Double microwave shielding

Bialkali molecules are lossy

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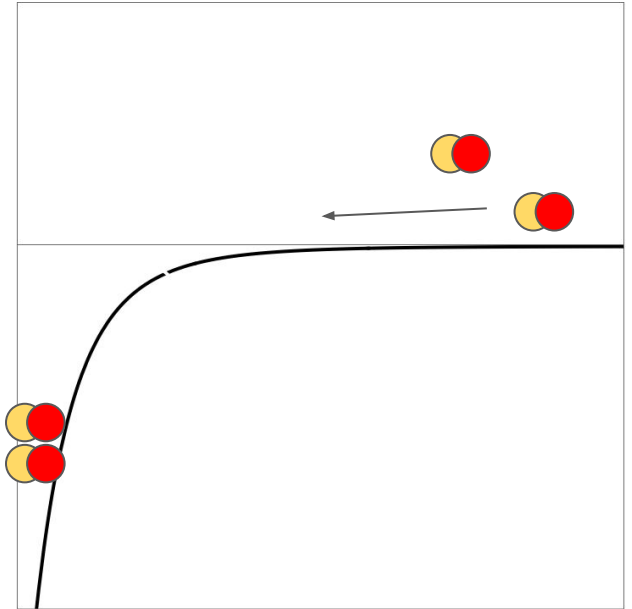
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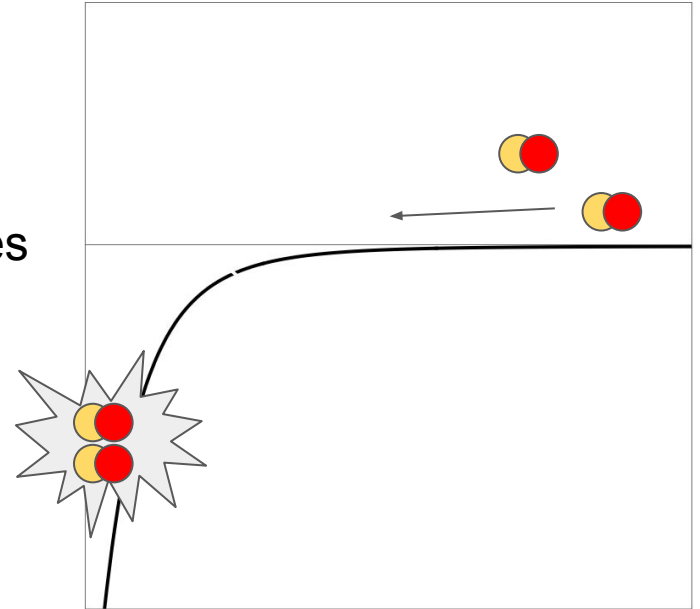
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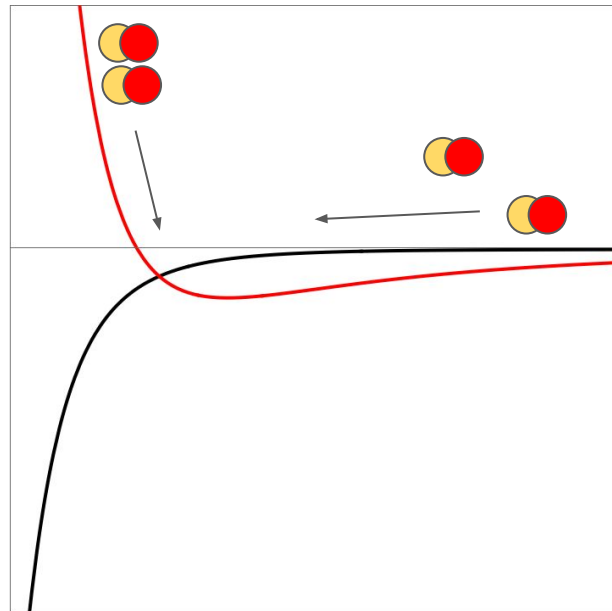
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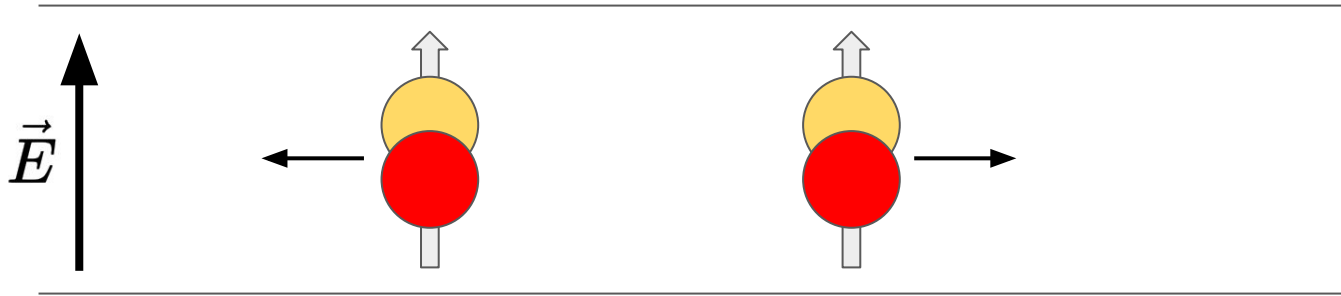
Collisional shielding:

Modify the intermolecular potential to prevent molecules from slamming into each other



Collisional shielding with dc E-field + trap confinement

Polarize dipoles with DC field + Suppress (attractive) head-to-tail collisions
→ generate mostly repulsive side-to-side interaction

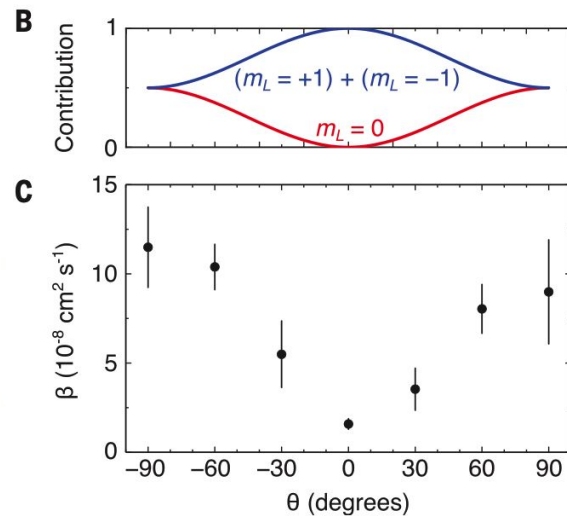
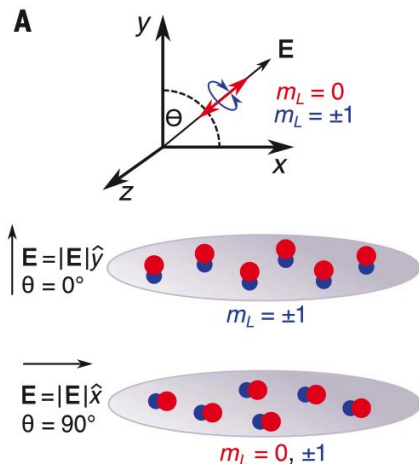
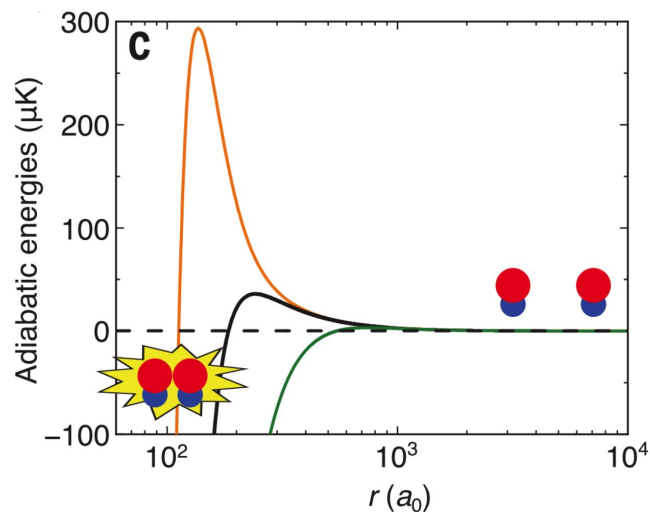


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K. Matsuda et al, Science **370** (2020) [1]



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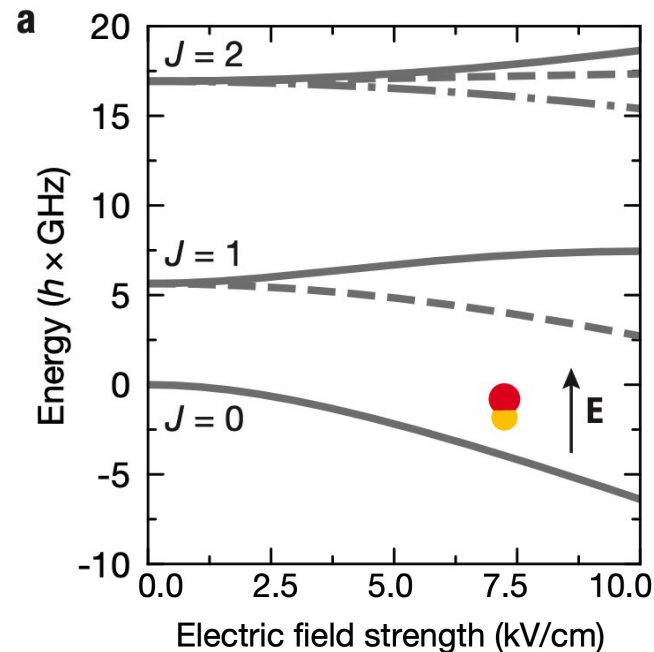
Some disadvantages

Strong E-fields required: ~ 7 kV/cm for NaK

Charging the vacuum chamber

→ uncontrolled shifts of molecule lines. [XYC \[2\]](#)

XYC, PhD thesis, LMU (2023) [2]



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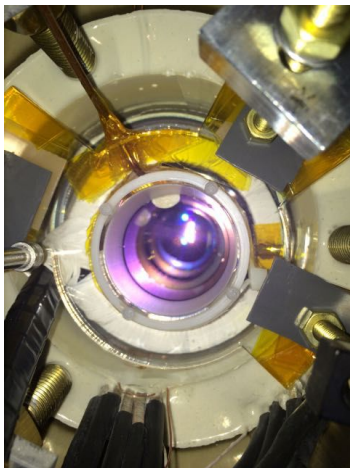
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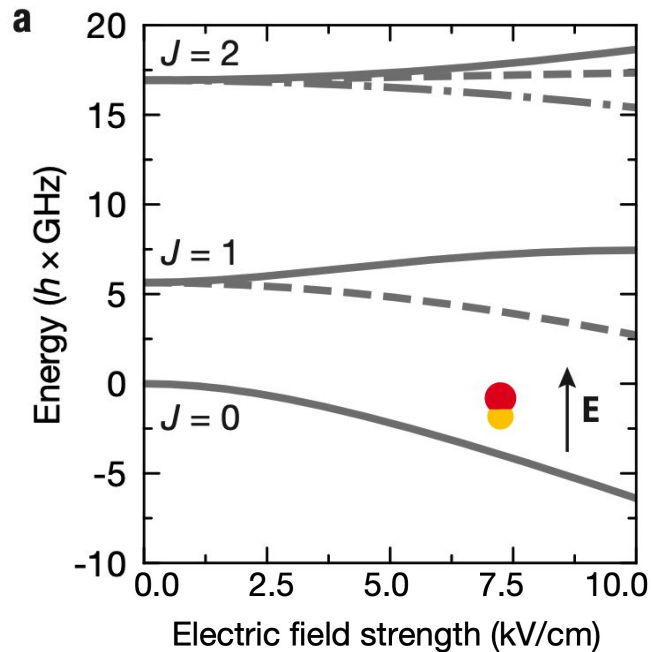
Limited tunability

HV electrode circa 2014

Fermi1 OneNote



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Collisional shielding with microwaves

σ^+ polarized microwave, blue-detuned from $J = 0 \rightarrow 1$ rotational transition

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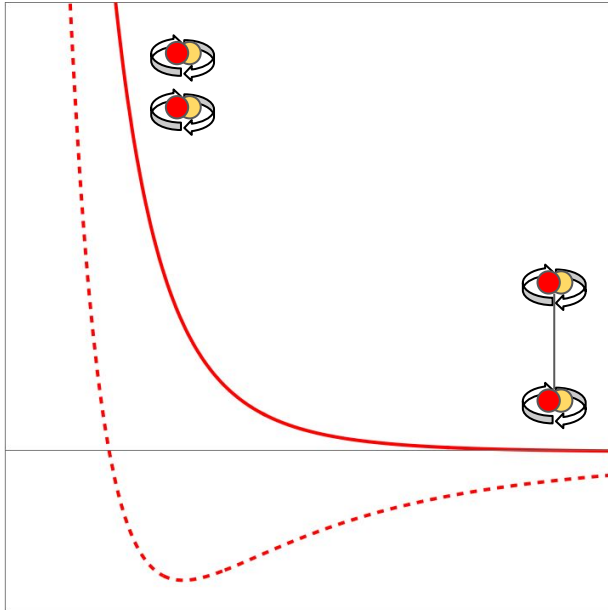
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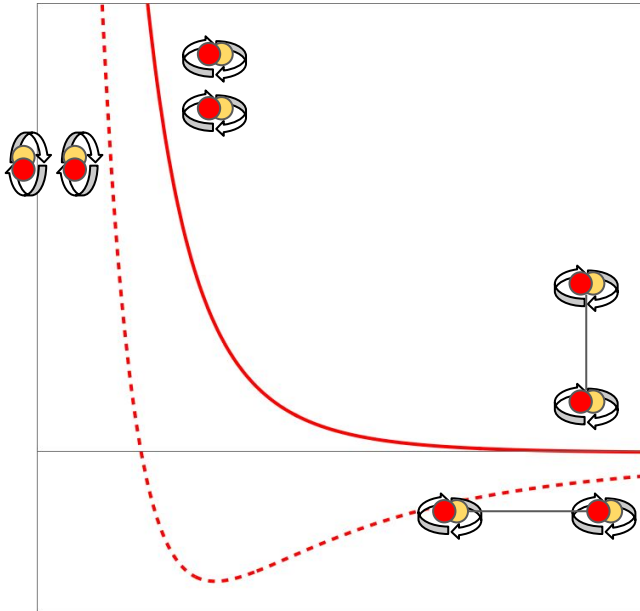


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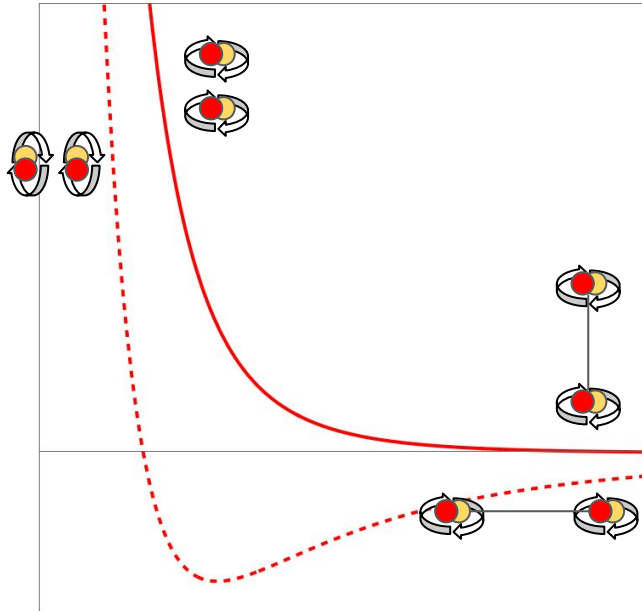
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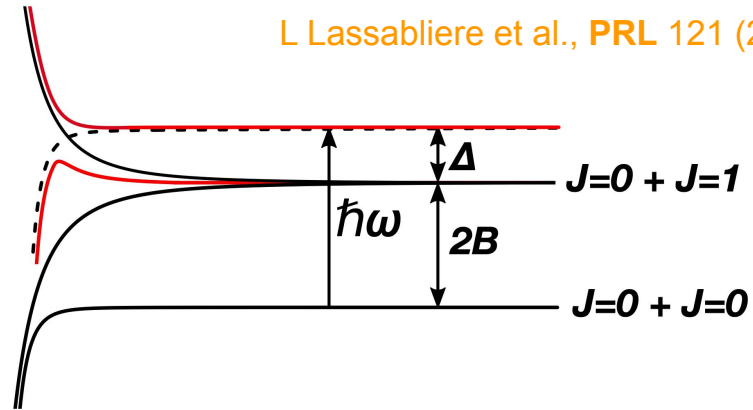
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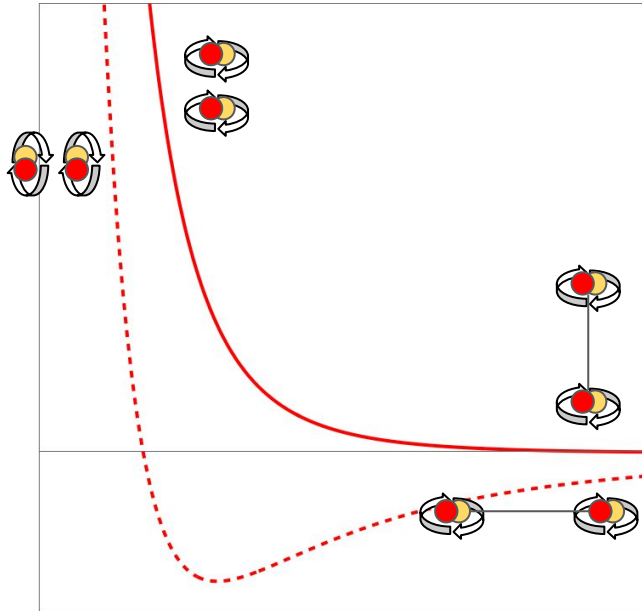
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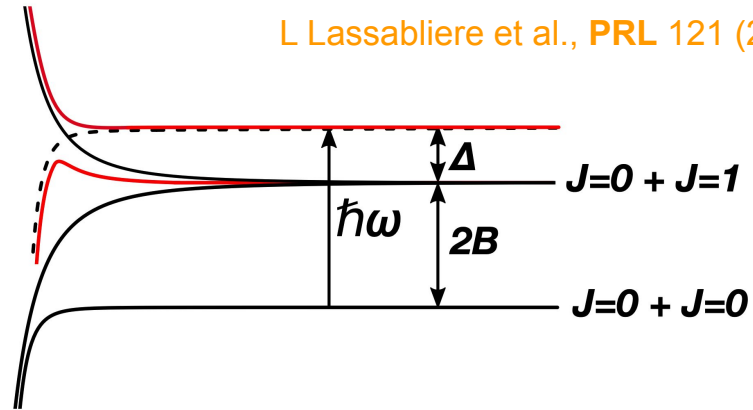
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Quantitative picture needs full QM treatment

Molecule in a microwave field

Model: Rigid rotor in an oscillatory field, ignore hfs

$$H = B_{\text{rot}} \mathbf{J}^2 - d_0 \hat{d} \cdot \vec{E}_{\text{MW}}$$

$\approx 2.7\text{D}$ for NaK, dipole moment in body-fixed frame

$$\vec{E}_{\text{MW}} = \frac{1}{2} E_{\text{MW}} e^{-i\omega t} (\hat{\mathbf{e}}_+ \cos \xi + \hat{\mathbf{e}}_- \sin \xi) + \text{c.c.}$$

Ellipticity parameter

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Rotating frame Hamiltonian

$$H = \frac{\hbar}{2} \begin{pmatrix} 0 & \Omega \sin \xi & 0 & \Omega \cos \xi \\ \Omega \sin \xi & -2\Delta & 0 & 0 \\ 0 & 0 & -2\Delta & 0 \\ \Omega \cos \xi & 0 & 0 & -2\Delta \end{pmatrix}$$

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$$d_0 \langle i | \sum_{m=-1}^1 C_{1,m} E_m^* | j \rangle$$

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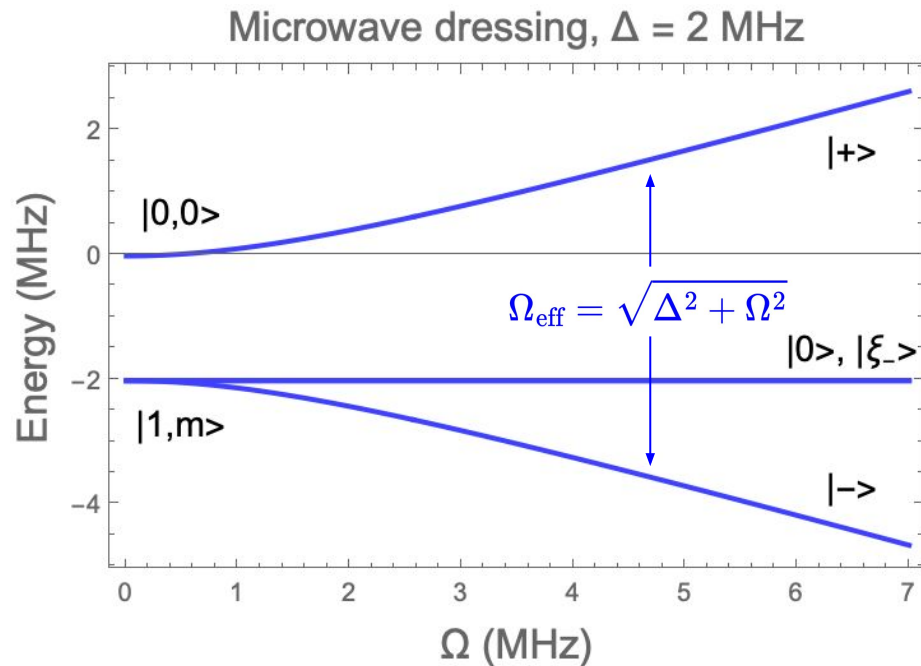
$$\Omega = d_0 E_{\text{MW}} / \hbar \sqrt{3}$$

$$\Delta = \omega - 2B_{\text{rot}} / \hbar$$

$\approx 2\pi \times 5.643 \text{ GHz}$ for NaK

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Microwave-dressed states



Adiabatically connected to $|0,0\rangle$

$$|+\rangle = +u |0,0\rangle + v \sin \xi |1,-1\rangle + v \cos \xi |1,1\rangle$$

$$|-\rangle = -v |0,0\rangle + u \sin \xi |1,-1\rangle + u \cos \xi |1,1\rangle$$

$$|\xi_{-}\rangle = \cos \xi |1,-1\rangle - \sin \xi |1,1\rangle$$

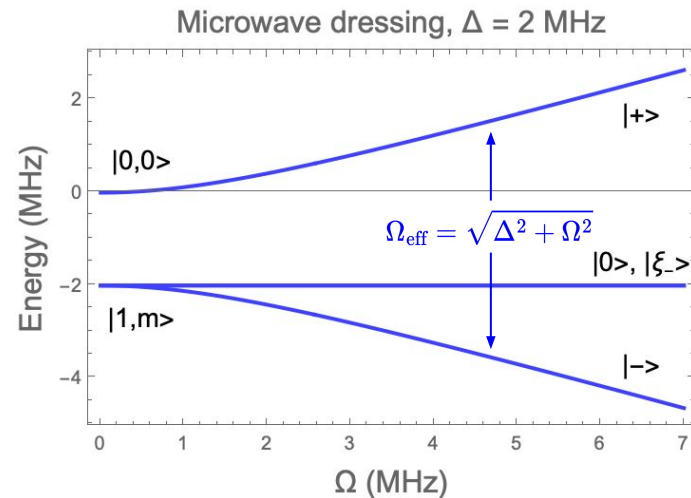
$$|0\rangle = |1,0\rangle$$

$$u = \sqrt{(1 + \Delta/\Omega_{\text{eff}})/2}$$

$$v = \sqrt{(1 - \Delta/\Omega_{\text{eff}})/2}$$

Dressed state lifetime

Phase noise in the microwave sets a bound on the (one-body) lifetime of $|+\rangle$

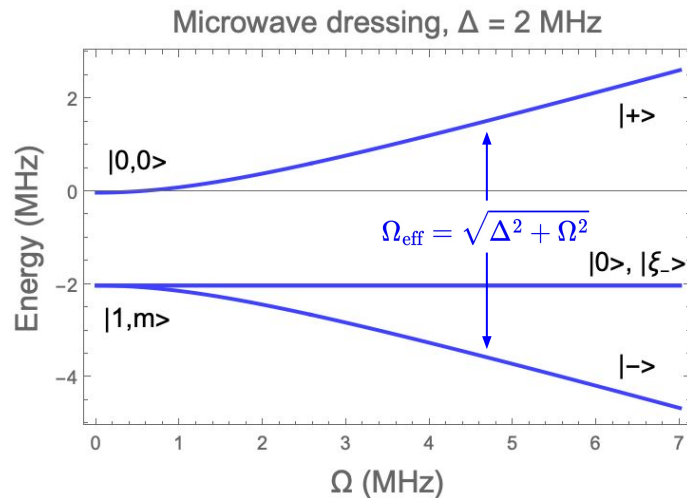


Dressed state lifetime

Phase noise in the microwave sets a bound on the (one-body) lifetime of (+)

Model [2] Phase fluctuation

$$\hat{h}_{\text{in}} = \hbar\Delta|0,0\rangle\langle 0,0| + \left(\frac{\hbar\Omega}{2} e^{-i\phi(t)} |\xi_+\rangle\langle 0,0| + \text{h.c.} \right)$$

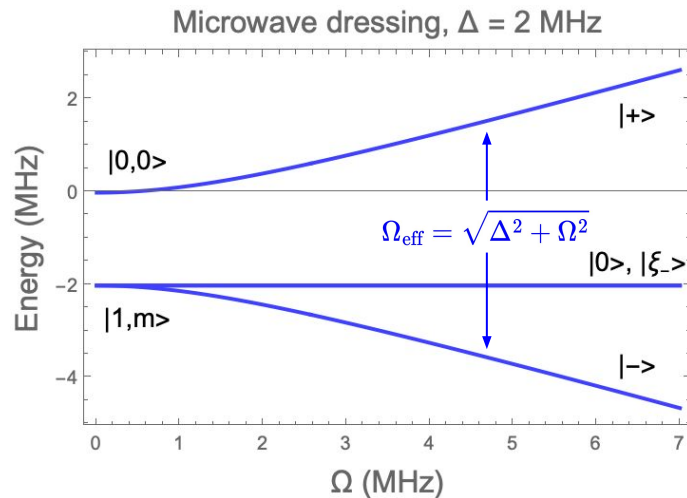


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Model [2] RWA, in dressed basis, assuming small modulations...

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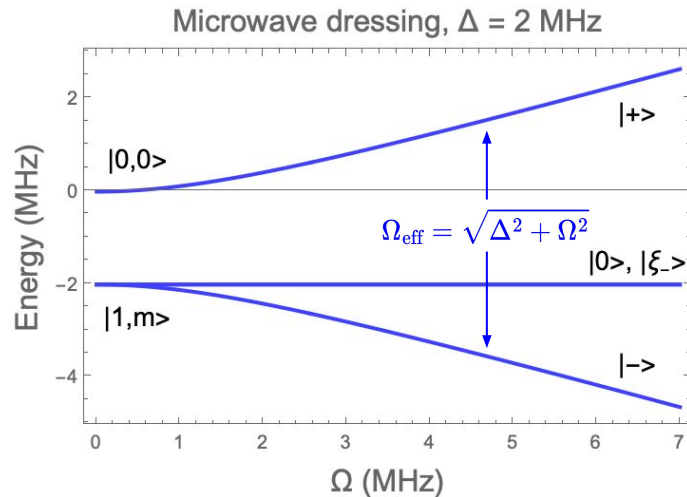
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$$P_{|+\rangle\rightarrow|-\rangle} = \lim_{T\rightarrow\infty} \frac{1}{T} \left| \frac{-i}{\hbar} \int_0^T \phi(t') \frac{\hbar\Omega}{2} e^{i\Omega_{\text{eff}}t'} dt' \right|^2$$



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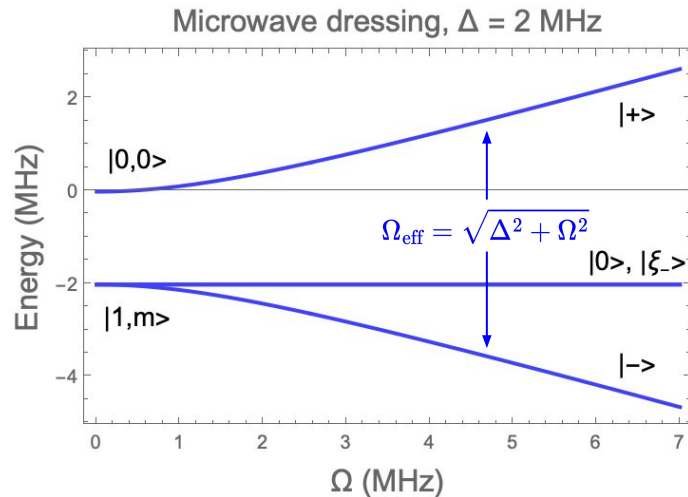
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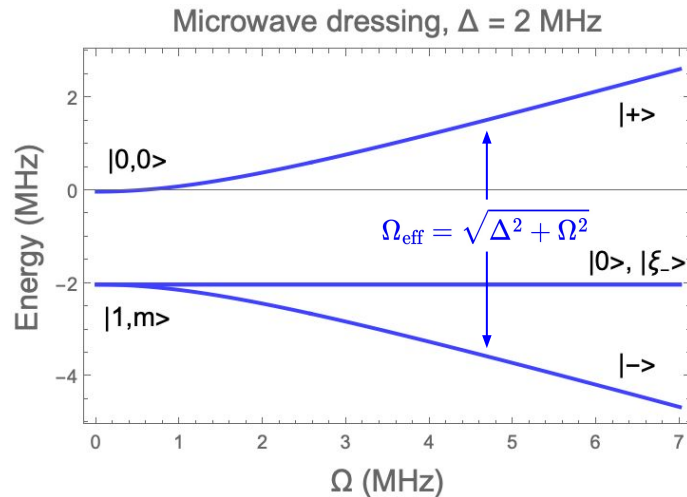
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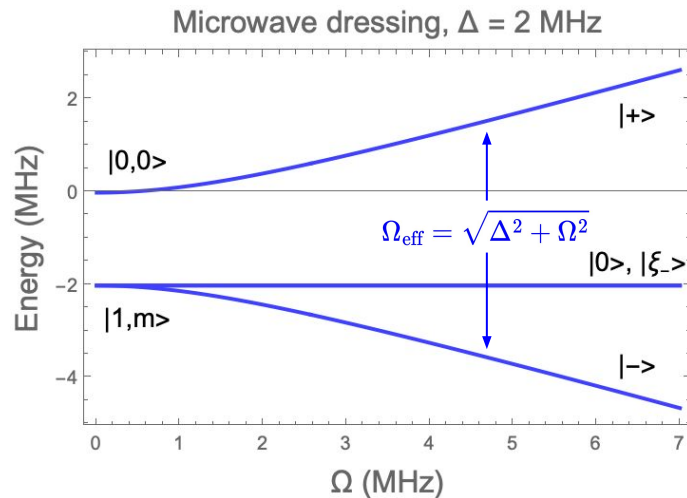
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Single-sideband phase noise at f_{eff}



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Exp. decay of (+) population

$$P_+(t) = \exp\left(-\frac{\Omega^2}{2}\mathcal{L}(f_{\text{eff}})t\right)$$

At $\Omega = 2\pi \times 10$ MHz, need -170 dBc/Hz to reach ~10s lifetime

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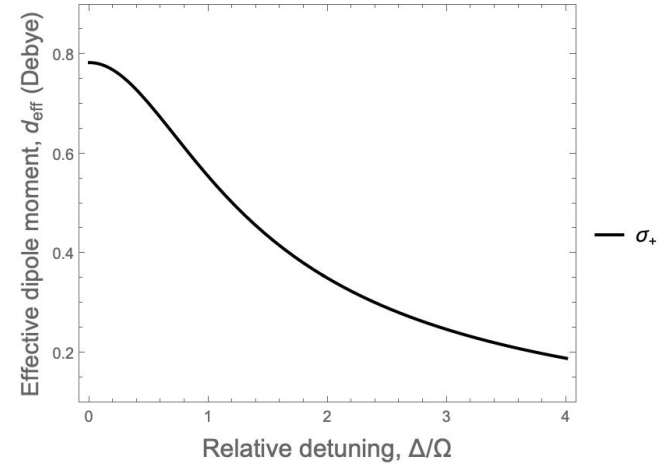
Clean MW source is crucial!



Two molecules in a microwave field

Induced/Effective dipole moment

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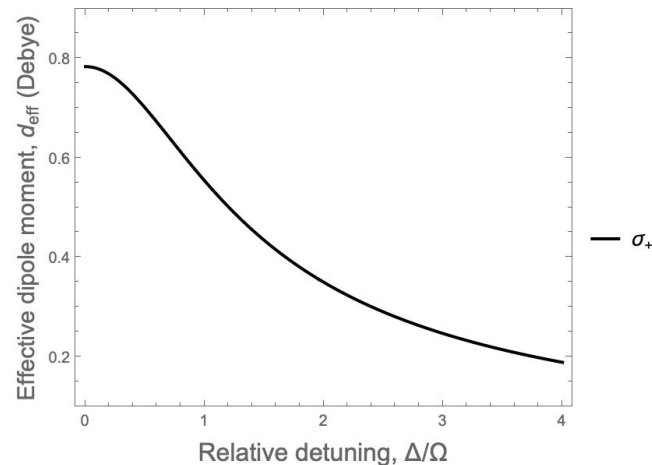
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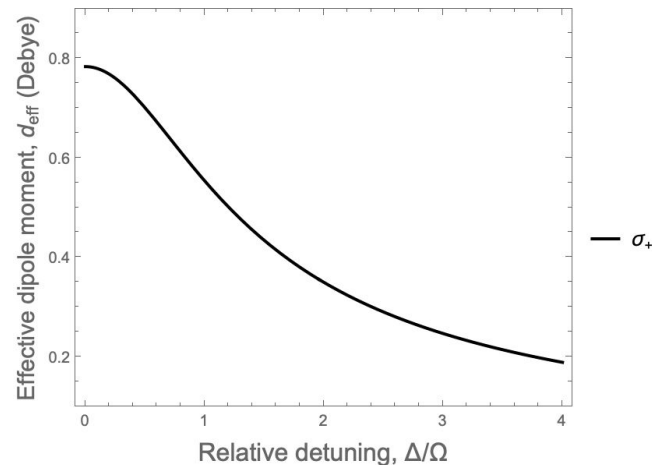
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→ First-order approximation, since V_{dd} is not diagonal in the dressed^{⊗2} basis



Two molecules in a microwave field

Full QM solution is a complicated multichannel scattering problem:

$$H_{2B,\text{full}} = K + H_{\text{mol-MW}}^{(1)} + H_{\text{mol-MW}}^{(2)} - \frac{C_6}{R^6} + \hat{V}_{\text{dd}}(\vec{R})$$

Kinetic terms for colliding molecules: radial + angular

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Assume kinetic terms $\ll \Omega$, can diagonalize H_{2B} with $K=0$ to find the microwave-dressed intermolecular potentials $\rightarrow$ identify shielding parameters

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 “Adiabatic potentials”

Adiabatic potentials

Two-body Hamiltonian, excluding kinetic terms

$$H_{2B} = H_{\text{mol-MW}}^{(1)} + H_{\text{mol-MW}}^{(2)} - \frac{C_6}{R^6} + \hat{V}_{\text{dd}}(\vec{R})$$

Dipole-dipole interaction operator

$$\hat{V}_{\text{dd}}(\vec{R}) = \frac{d_0^2}{4\pi\epsilon_0 R^3} \left\{ \hat{d}_1 \cdot \hat{d}_2 - 3(\hat{d}_1 \cdot \hat{r})(\hat{d}_2 \cdot \hat{r}) \right\}$$

Single-molecule basis: $\{|+\rangle, |-\rangle, |0\rangle, |\xi_{-1}\rangle\}$ $\rightarrow$ diagonalizes $H_{\text{mol-MW}}^{(i)}$

Two-molecule basis is 7-dimensional by symmetry and “darkness”

$$|++\rangle, | +0 \rangle_s, | +\xi_{-} \rangle_s, | +- \rangle_s, | -0 \rangle_s, | -\xi_{-} \rangle_s, | -- \rangle$$

Matrix elements of H_{2B}

Since the kets encode products of spherical harmonics, it is more convenient to express the dipole-dipole operator in similar terms:

$$\hat{V}_{dd}(\vec{R}) = -8\sqrt{\frac{2}{15}}\pi^{3/2}\frac{d_0^2}{4\pi\epsilon_0 R^3}\sum_{m=-2}^2 Y_{2,m}^*(\hat{R})\Sigma_{2,m}$$

Rank-2 spherical tensor

Direction of dipole j

$$\Sigma_{2,0} = \frac{1}{\sqrt{6}}(\hat{d}_1^+ \hat{d}_2^- + \hat{d}_1^- \hat{d}_2^+ + 2\hat{d}_1^0 \hat{d}_2^0)$$

$$\Sigma_{2,\pm 1} = \frac{1}{\sqrt{2}}(\hat{d}_1^\pm \hat{d}_2^0 + \hat{d}_1^0 \hat{d}_2^\pm) \quad \Sigma_{2,\pm 2} = \hat{d}_1^\pm \hat{d}_2^\pm$$

$$d_j^\pm = Y_{1,\pm 1}(\hat{\mathbf{d}}_j)$$

$$d_j^0 = Y_{1,0}(\hat{\mathbf{d}}_j)$$

Matrix elements of H_{2B}

Provided in [F. Deng et al PRL 130 \(2023\) \[4\]](#). Can also find after a few Mathematica cells.

$$\hat{h}_{in}(1) + \hat{h}_{in}(2) = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2}(\delta - \Omega_{\text{eff}}) & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2}(\delta - \Omega_{\text{eff}}) & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\Omega_{\text{eff}} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2}(\delta - 3\Omega_{\text{eff}}) & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2}(\delta - 3\Omega_{\text{eff}}) & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -2\Omega_{\text{eff}} \end{pmatrix},$$

$$\Sigma_{2,0} = \frac{1}{4\pi\sqrt{6}} \begin{pmatrix} -2u^2v^2 & 0 & 0 & -\sqrt{2}uv(u^2 - v^2) & 0 & 0 & 2u^2v^2 \\ 0 & 2u^2 & 0 & 0 & -2uv & 0 & 0 \\ 0 & 0 & -u^2 & 0 & 0 & uv & 0 \\ -\sqrt{2}uv(u^2 - v^2) & 0 & 0 & -(u^2 - v^2)^2 & 0 & 0 & \sqrt{2}uv(u^2 - v^2) \\ 0 & -2uv & 0 & 0 & 2v^2 & 0 & 0 \\ 0 & 0 & uv & 0 & 0 & -v^2 & 0 \\ 2u^2v^2 & 0 & 0 & \sqrt{2}uv(u^2 - v^2) & 0 & 0 & -2u^2v^2 \end{pmatrix}$$

$$\Sigma_{2,1} = \frac{1}{4\pi\sqrt{2}} \begin{pmatrix} 0 & \sqrt{2}u^2v & 0 & 0 & -\sqrt{2}uv^2 & 0 & 0 \\ 0 & 0 & -u^2 & 0 & 0 & uv & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & u(u^2 - v^2) & 0 & 0 & -v(u^2 - v^2) & 0 & 0 \\ 0 & 0 & uv & 0 & 0 & -v^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\sqrt{2}u^2v & 0 & 0 & \sqrt{2}uv^2 & 0 & 0 \end{pmatrix},$$

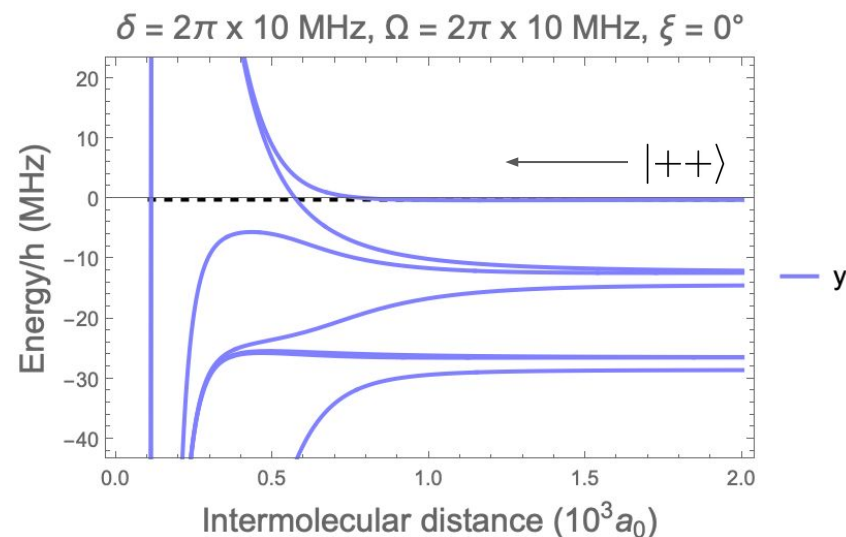
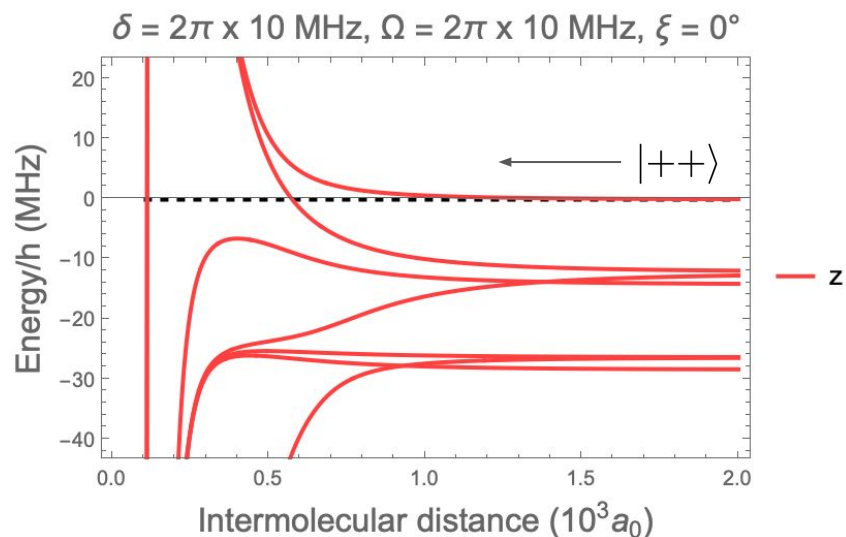
$$\Sigma_{2,2} = -\frac{1}{4\pi} \begin{pmatrix} 0 & 0 & \sqrt{2}u^2v & 0 & 0 & -\sqrt{2}uv^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & u(u^2 - v^2) & 0 & 0 & -v(u^2 - v^2) & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\sqrt{2}u^2v & 0 & 0 & \sqrt{2}uv^2 & 0 \end{pmatrix}.$$

```
basis = {S1, S2, S3, S4, S5, S6, S7};
In[458]:= SphericalHarmony[q3_, q2_, q1_] := Sqrt[(2 q1[[1]] + 1) * (2 q2[[1]] + 1) / (4 Pi * (2 q3[[1]] + 1))] *
ClebschGordan[{q1[[1]], 0}, {q2[[1]], 0}, {q3[[1]], 0}] *
ClebschGordan[{q1[[1]], q1[[2]]}, {q2[[1]], q2[[2]]}, {q3[[1]], q3[[2]]}];
LimitTotalJFinal[qn1_, qn2_] := Boole[qn1[[1]] + qn2[[1]] < 2];
InnerProd[braComponents1_, braComponents2_, op1_, op2_, ketComponents1_, ketComponents2_] :=
Sum[
Sum[LimitTotalJFinal[quantumNumbers[[i]], quantumNumbers[[k]] *
LimitTotalJFinal[quantumNumbers[[j]], quantumNumbers[[l]] * braComponents1[[i]] * ketComponents1[[j]] *
SphericalHarmony[quantumNumbers[[i]], op1, quantumNumbers[[j]], {i, 1, 4}, {j, 1, 4}] *
braComponents2[[k]] * ketComponents2[[l]] * SphericalHarmony[quantumNumbers[[k]], op2, quantumNumbers[[l]],
{k, 1, 4}, {l, 1, 4}];
MatrixElement[op_, bra_, ket_] :=
(1 / Sqrt[Dimensions[bra][1]] * (1 / Sqrt[Dimensions[ket][1]])) *
Sum[
Sum[InnerProd[bra[[b]][1], bra[[b]][2], op[[i]][1], op[[i]][2], ket[[k]][1], ket[[k]][2]], {i, 1, Dimensions[op][1]],
{b, 1, Dimensions[bra][1]}, {k, 1, Dimensions[ket][1]}] // FullSimplify;
In[459]:= S20Matrix = Table[4 * Pi * MatrixElement[S20, basis[[r]], basis[[c]], {r, 1, Dimensions[basis][1]},
{c, 1, Dimensions[basis][1]}];
S2p1Matrix = Table[4 * Pi * MatrixElement[S2p1, basis[[r]], basis[[c]], {r, 1, Dimensions[basis][1]},
{c, 1, Dimensions[basis][1]}];
S2m1Matrix = Table[4 * Pi * MatrixElement[S2m1, basis[[r]], basis[[c]], {r, 1, Dimensions[basis][1]},
{c, 1, Dimensions[basis][1]}];
S2p2Matrix = Table[4 * Pi * MatrixElement[S2p2, basis[[r]], basis[[c]], {r, 1, Dimensions[basis][1]},
{c, 1, Dimensions[basis][1]}];
S2m2Matrix = Table[4 * Pi * MatrixElement[S2m2, basis[[r]], basis[[c]], {r, 1, Dimensions[basis][1]},
{c, 1, Dimensions[basis][1]}];
```

An application of AMO1 Pset6 “Spherical Harmony”

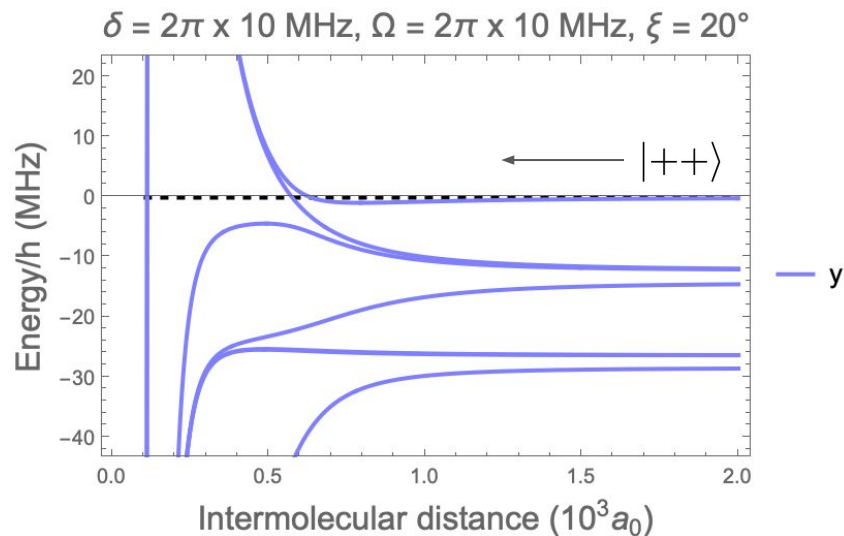
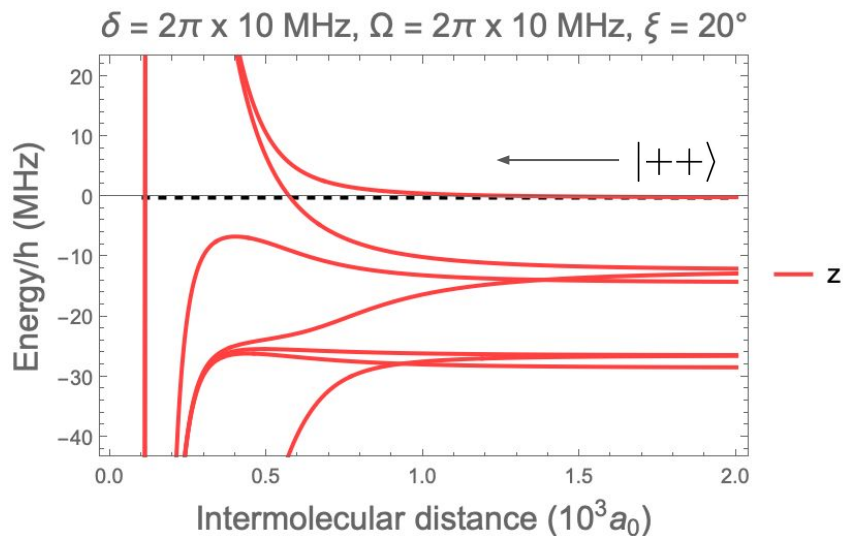
Adiabatic potentials for NaK, $\xi = 0^\circ$

MW-dressed molecules repel at short-range in all directions $\rightarrow$ shielded



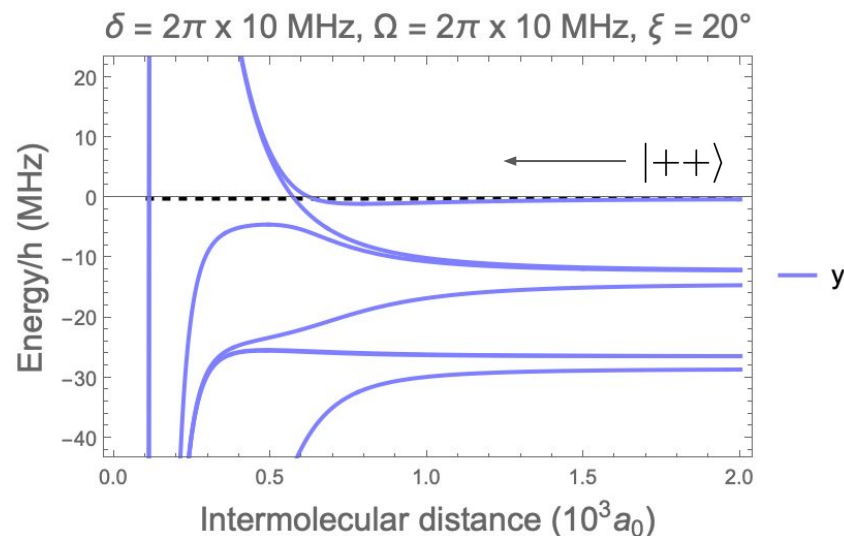
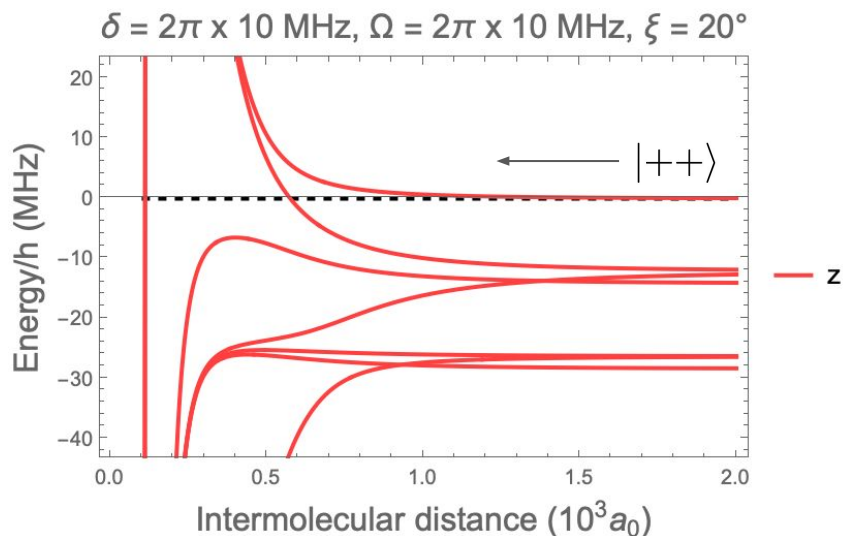
Role of ellipticity angle ξ

Interaction potential is most attractive along the direction of the MW ellipse



Role of ellipticity angle ξ

Interaction potential is most attractive along the direction of the MW ellipse

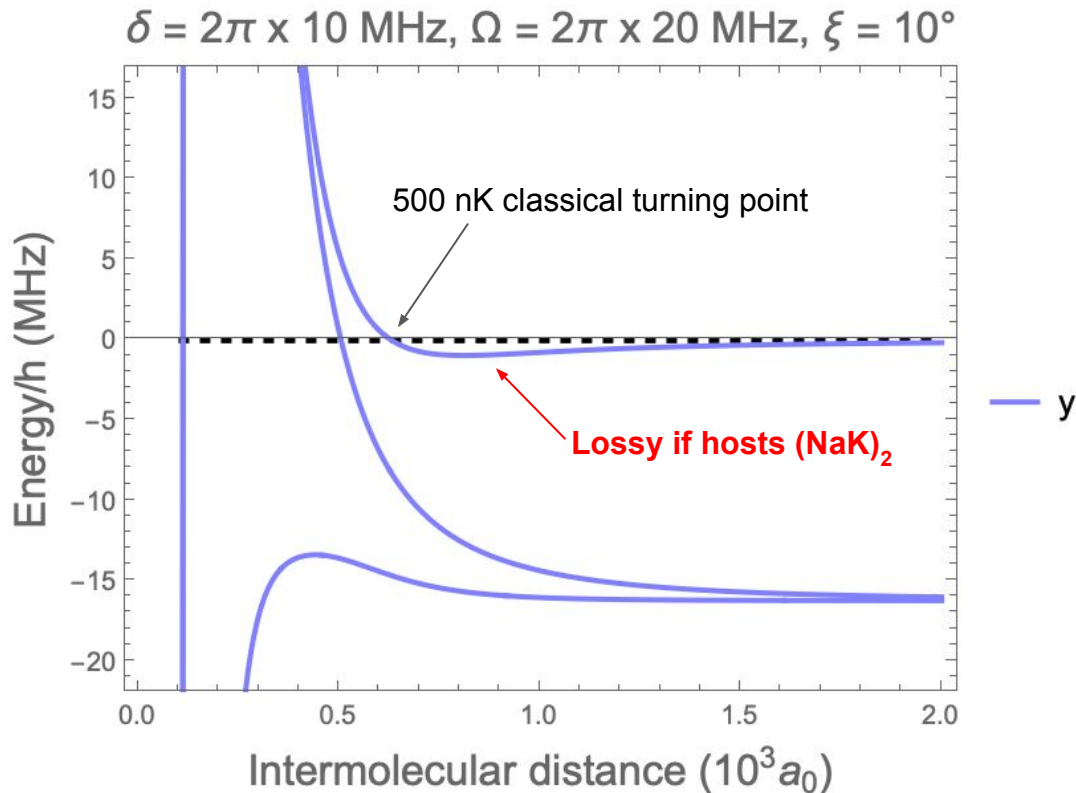


→ By tuning ξ , it is possible to create field-linked bound states $(\text{NaK})_2$ [5]

Optimal shielding parameters

Loss mechanisms:

- Molecules entering short range
- Tunneling to lower channels
- Forming lossy complexes
- Limited by worst collision axis



Optimal shielding parameters

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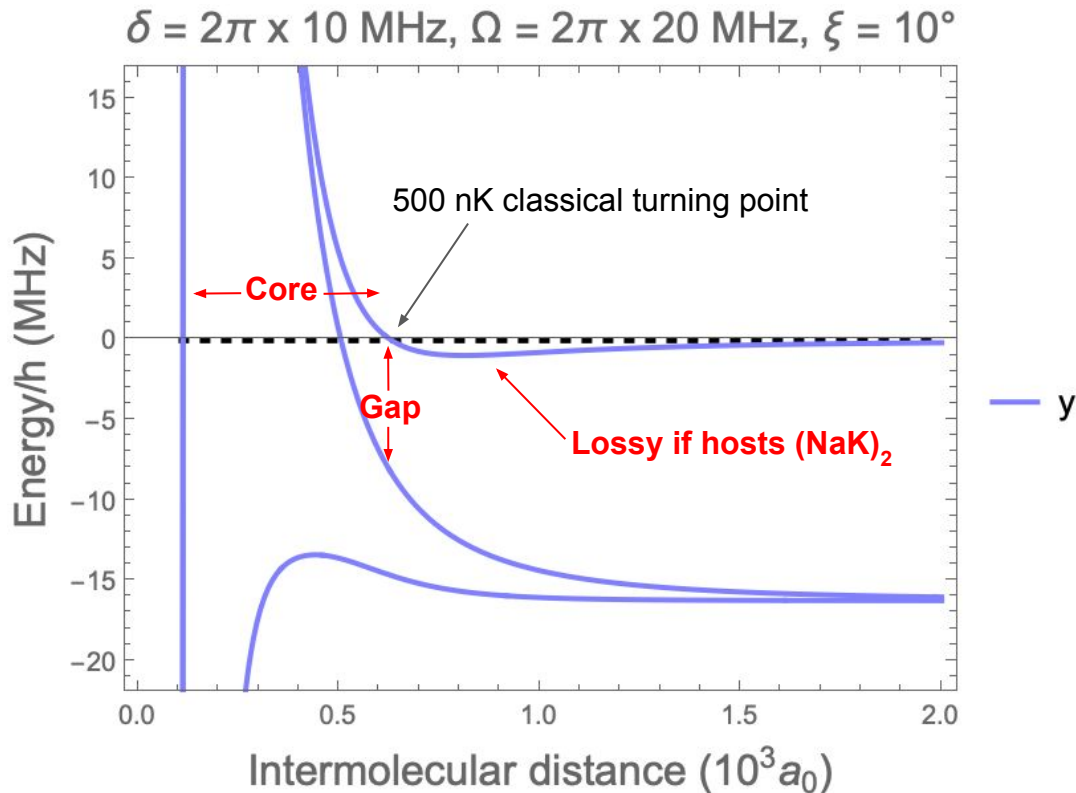
Requirements:

- Large shielding core
- Large shielding gap
- Low loss***

Trade-offs:

- $\Delta \uparrow, \Omega \uparrow = \text{Core} \downarrow \text{Gap} \uparrow$
- $\Omega \uparrow = \text{deeper well}^{***}$

Best to optimize by minimizing loss



Effective potential

It is convenient to have an analytic form for the uppermost adiabatic potential V^+

Generally, V^+ gets most contributions from the first three basis states

→ Can truncate H_{2B} to a 3x3, diagonalize, and get the highest eigenvalue

Result for small ξ :

$$C_3 = d^2 / [24\pi\epsilon_0(1 + \delta_r^2)]$$

$$C_6 = d^4 / [1120\pi^2\epsilon_0^2\Omega(1 + \delta_r^2)^{3/2}]$$

$$\delta_r = \Delta/\Omega$$

$$V_{\text{eff}}(\xi, \vec{R}) = \frac{C_3}{2R^3} (3 \cos^2 \theta - 1 + 3 \sin 2\xi \sin^2 \theta \cos 2\phi) \\ + \frac{35C_6}{4R^6} \sin^2 \theta (\cos^2 \theta + 1 - 2 \sin 2\xi \cos^2 \theta \cos 2\phi - \sin^2 2\xi \sin^2 \theta \cos^2 2\phi)$$

Bound states revisited

Long-range attractive in xy plane

Short-range repulsive

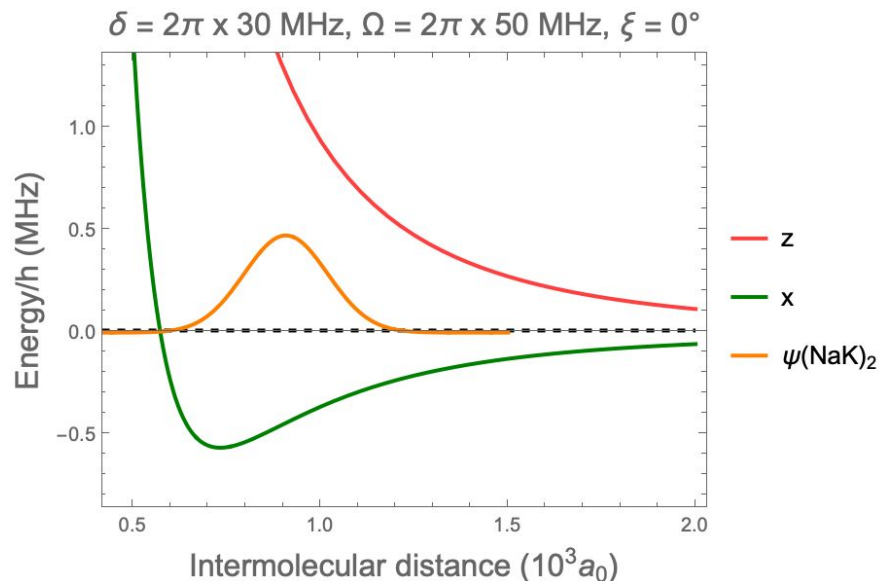
Effective potential:
$$V_{\text{eff}}(\xi, \vec{R}) = \frac{C_3}{2R^3} (3 \cos^2 \theta - 1 + 3 \sin 2\xi \sin^2 \theta \cos 2\phi) + \frac{35C_6}{4R^6} \sin^2 \theta (\cos^2 \theta + 1 - 2 \sin 2\xi \cos^2 \theta \cos 2\phi - \sin^2 2\xi \sin^2 \theta \cos^2 2\phi)$$

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Intermolecular potential is NOT repulsive in all direction $\rightarrow$ limitation of σ^+ dressing

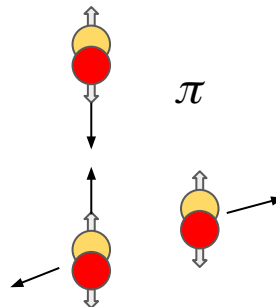
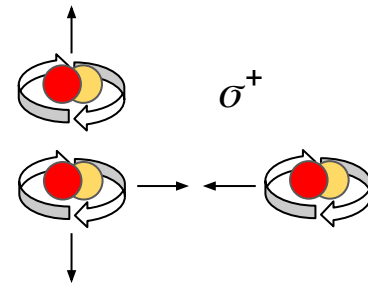
Two-body loss for strong σ^+ dressing
Limits shielding core/gap

Double microwave shielding [6]

Idea: Cancel C_3 term so the intermolecular interaction is repulsive in all directions

“First-order” dipole-dipole interaction, classical picture

$$V_{\text{eff}}(\xi, \vec{R}) = \frac{C_3}{2R^3} (3 \cos^2 \theta - 1 + 3 \sin 2\xi \sin^2 \theta \cos 2\phi) + \frac{35C_6}{4R^6} \sin^2 \theta (\cos^2 \theta + 1 - 2 \sin 2\xi \cos^2 \theta \cos 2\phi - \sin^2 2\xi \sin^2 \theta \cos^2 2\phi)$$



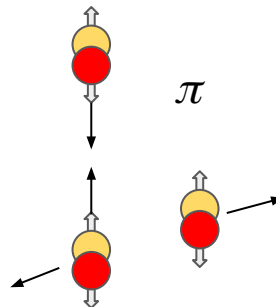
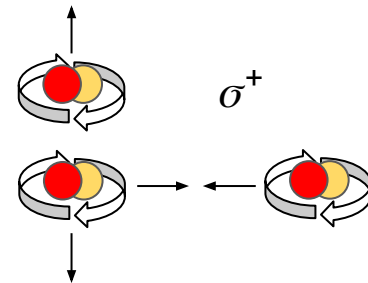
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→ Introduce an additional π -polarized MW
Add a DDI term with opposite sign



Molecule in a π and σ^+ microwave field

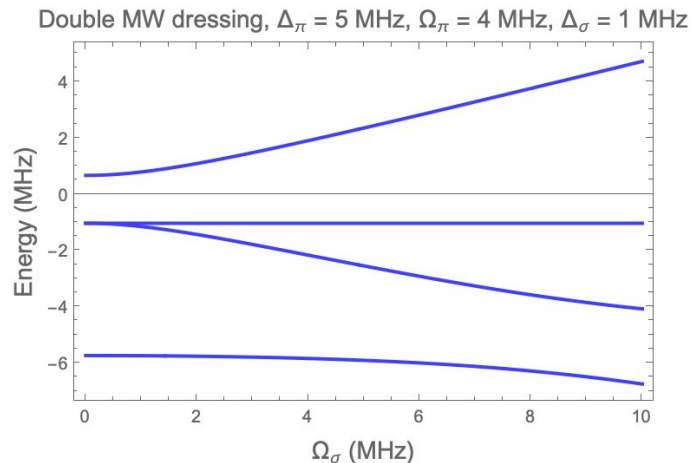
Dressed Hamiltonian

$$H_{\pi-\sigma^+} = \frac{\hbar}{2} \begin{pmatrix} 0 & 0 & \Omega_\pi & \Omega_\sigma \\ 0 & -2\Delta_\sigma & 0 & 0 \\ \Omega_\pi & 0 & -2\Delta_\pi & 0 \\ \Omega_\sigma & 0 & 0 & -2\Delta_\sigma \end{pmatrix}$$

An approximation, since there are many nearby bare channels

No closed-forms for dressed states or energies

Dressed states are parameterized not just by relative detunings (Δ/Ω) but also relative strengths of the MW fields (Ω_π/Ω_σ)

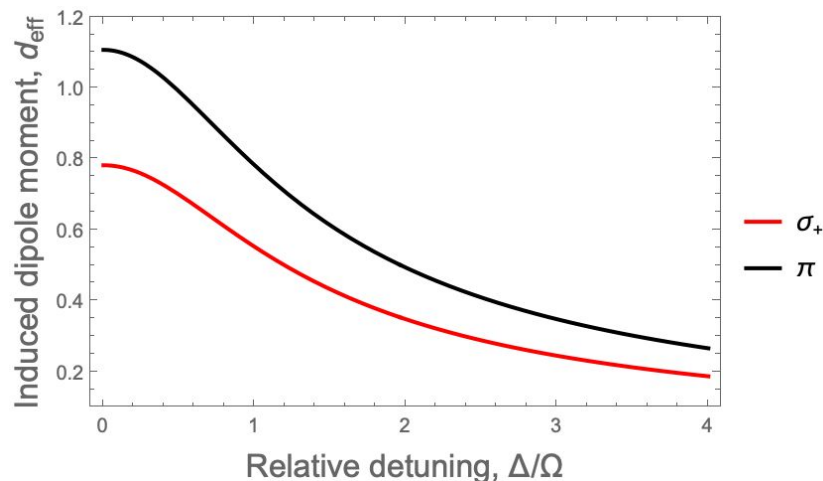


Molecule in a π and σ^+ microwave field

For pure π/σ^+ , the effective dipole moment d_{eff} is fully characterized by its magnitude:

$$d_{\text{eff},\sigma^+} = \frac{d_0}{\sqrt{12(1 + \Delta_{\sigma^+}^2/\Omega_{\sigma^+}^2)}}$$

$$d_{\text{eff},\pi} = \frac{d_0}{\sqrt{6(1 + \Delta_{\pi}^2/\Omega_{\pi}^2)}}$$



For combined π and σ^+ , the d_{eff} 's add in quadrature and real-valuedness matters
→ Need to consider the sign of the dipole-dipole interaction $\sim (d_{\text{eff}})^2$

Two molecules in a π and σ^+ microwave field

DDI is attractive for σ^+ and repulsive for π : $\langle ++ | \hat{V}_{\text{DDI}} | ++ \rangle = \frac{2d^2 P_2(\cos \theta)}{4\pi\epsilon_0 R^3} \begin{cases} -1/12 & \text{for } \sigma^+ \\ 1/6 & \text{for } \pi \end{cases}$

Define d_{eff} for σ^+ as *imaginary*: $d_{\text{eff},\sigma^+} = \frac{id_0}{\sqrt{12(1 + \Delta_{\sigma^+}^2/\Omega_{\sigma^+}^2)}}$

Induced dipole moment: $d_{\text{eff}} = \sqrt{|d_\pi|^2 - |d_{\sigma^+}|^2}$

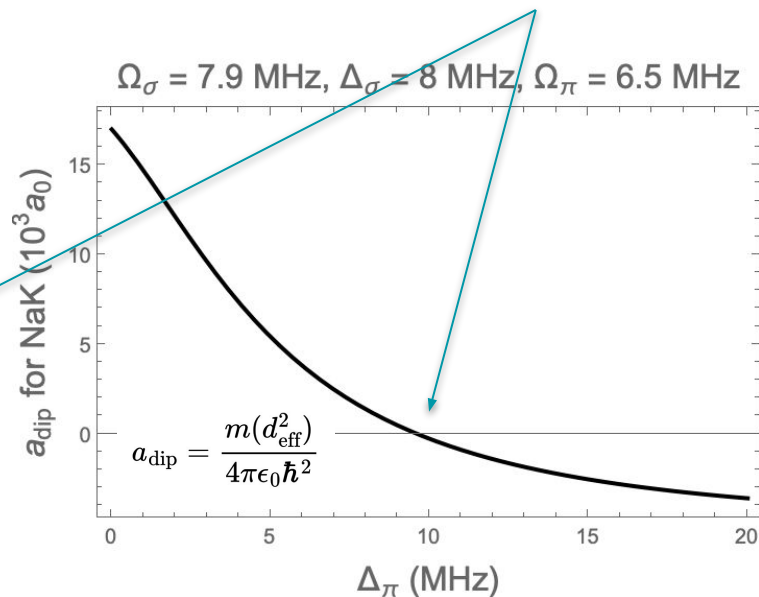
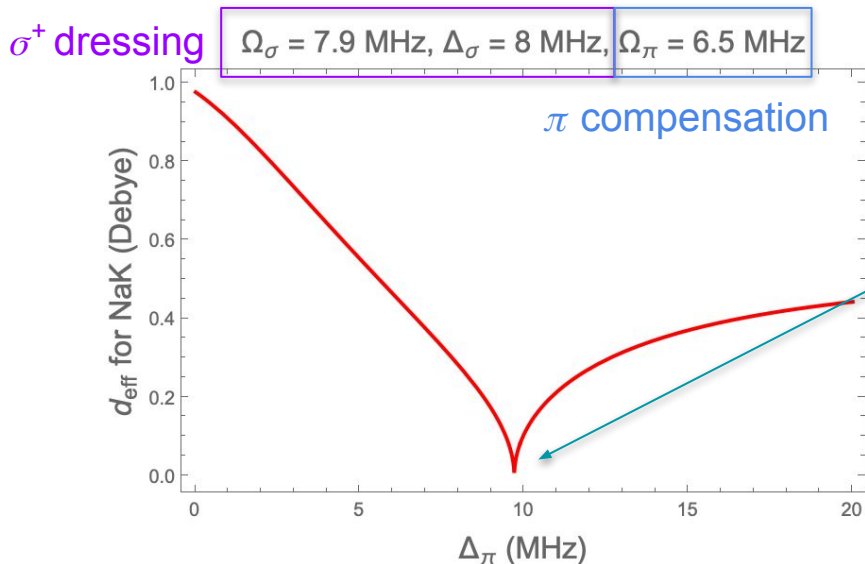
Define dipolar length, analogue of scattering length: $a_{\text{dip}} = \frac{m(d_{\text{eff}}^2)}{4\pi\epsilon_0 \hbar^2}$

$\rightarrow a_{\text{dip}}$ is widely tunable across zero by appropriate combinations of $\{\Omega_{\sigma^+}, \Omega_\pi, \Delta_{\sigma^+}, \Delta_\pi\}$

Two molecules in a π and σ^+ microwave field

Typically, near-optimal σ^+ dressing parameters are set.
Then, π dressing is optimized to remove first-order DDI.

DDI $\sim a_{\text{dip}}$ vanishes
“Compensation point”



Effective potential with π compensation

Full quantum scattering / adiabatic potential treatment is beyond this talk.

However, it suffices to revisit the **effective potential** idea.

For σ^+ dressing:

Set by field strength(s) Ω and detuning(s) Δ

$$V_{\text{eff}}(\vec{R}) = \frac{C_3}{R^3}(3 \cos^2 \theta - 1) + \frac{35C_6}{4R^6} \sin^2 \theta (\cos^2 \theta + 1)$$

$C_3 = d^2/[24\pi\epsilon_0(1 + \delta_r^2)]$
 $C_6 = d^4/[1120\pi^2\epsilon_0^2\Omega(1 + \delta_r^2)^{3/2}]$
 $\delta_r = \Delta/\Omega$

For double microwave dressing [7]:

$$V_{\text{eff}}(\vec{R}) = \frac{C_3}{R^3}(3 \cos^2 \theta - 1) + \frac{C_6}{R^6} \left[\sin^4 \theta + \frac{w_1}{w_2} \sin^2 \theta \cos^2 \theta + \frac{w_0}{w_2} (3 \cos^2 \theta - 1)^2 \right]$$

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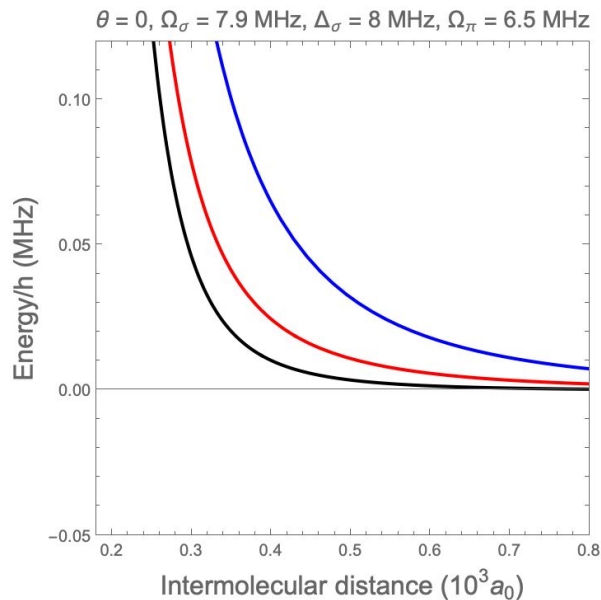
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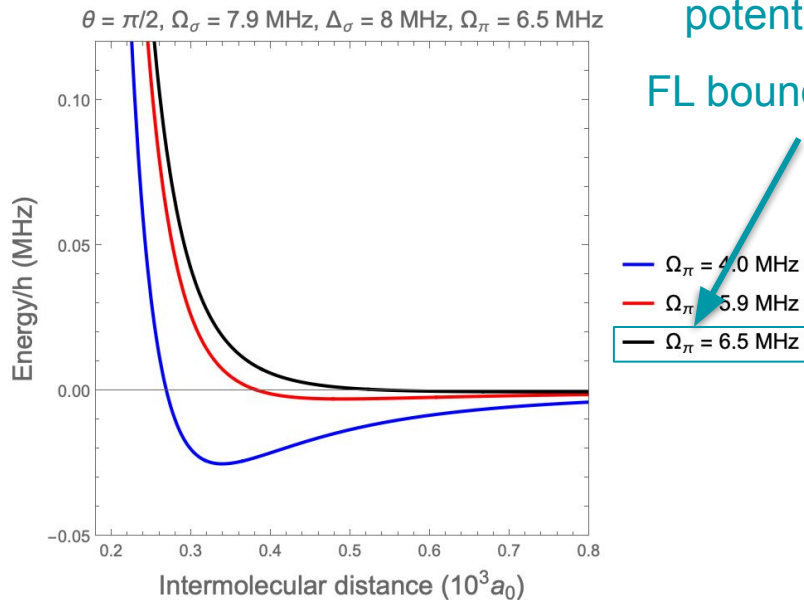
$$V_{\text{eff}}(\vec{R}) = \frac{C_3}{R^3}(3 \cos^2 \theta - 1) + \frac{C_6}{R^6} \left[\sin^4 \theta + \frac{w_1}{w_2} \sin^2 \theta \cos^2 \theta + \frac{w_0}{w_2} (3 \cos^2 \theta - 1)^2 \right]$$

Effective potential with double microwave shielding

Collisions along z



Collisions along x



Repulsive $1/R^6$ effective potential in **all** directions

FL bound states cannot exist

References

- [1] Kyle Matsuda et al., Science **370** (2020)
- [2] Chen Xingyan, PhD Dissertation, LMU (2023)
- [3] L Lassabliere et al., PRL **121** (2018)
- [4] F. Deng et al., PRL **130**.18 (2023)
- [5] XY Chen et al., Nature 614.7946 (2023)
- [6] K. Tijs et al. *arXiv:2501.08095* (2025)
- [7] F. Deng et al. *arXiv:2501.05210* (2025).